Project Report: Retail Sales Performance Analysis

Project Details

- **Project Title:** Retail Sales Performance Analysis
- **Project ID:** DA-011
- **Project Domain:** Retail & Sales
- **Current Status:** Completed (Analysis, Visualizations & Insights Finalized)

Project Description

This project looks at store sales data to help businesses make smart choices. It focuses on three main areas:

- **Sales Trends:** Track how much money the store makes each day, month, and 3-month period.
- **Best and Worst Products:** Find out which items sell fast and which ones sell slowly.
- **Customer Shopping Habits:** See how often customers shop, how much they buy at one time, and how they pay.

Methodology & System Architecture

1. Step-by-Step Project Approach

To ensure our findings are reliable and reproducible, our team followed a practical 5-step pipeline:

```
[Raw CSV Data]
       |
       ▼
```

```
[Step 1: Data Cleaning] ———> Remove nulls, fix date formats, handle duplicate rows
       |
       ▼
[Step 2: Metric Computation] -> Calculate Gross Revenue, Order Volumes, Average Bill
       |
       ▼
[Step 3: Visual Analytics] ——> Generate time-series plots, category bar charts
       |
       ▼
[Step 4: Business Insights] —-> Suggest inventory changes & promotion timing
```

1. **Data Collection & Ingestion:** We load transaction records stored in CSV format into Python using Pandas dataframes.
2. **Data Cleaning & Validation:** Raw records usually have empty cells, wrong data types, or duplicate bills. We validate every column before calculating anything.
3. **Exploratory Data Analysis (EDA):** We group records by month, category, and store location to see patterns that are invisible in raw spreadsheets.
4. **Data Visualization:** We convert raw numbers into readable visual charts (bar charts, line graphs) so non-technical managers can understand them immediately.
5. **Actionable Recommendations:** We conclude with clear suggestions, such as which products to restock before peak sales periods.

Technology Used

Technology / Tool	Type	Description
Python	Programming Language	The core language that powers the entire data pipeline and handles code execution.

Technology / Tool	Type	Description
Jupyter Notebook	Interactive Environment	An open-source web application used for creating and sharing documents that contain live code, equations, visualizations, and narrative text.
Google Colab	Cloud Notebook Environment	A hosted Jupyter notebook service provided by Google that runs entirely in the cloud with free access to computing resources.
Pandas	Python Library	Used for loading, cleaning, and manipulating tabular data (similar to Excel spreadsheets).
NumPy	Python Library	Handles large arrays of numbers and performs fast mathematical calculations behind the scenes.
Matplotlib	Visualization Libraries	Used to generate charts, graphs, and visual reports to easily spot business trends.
CSV	File Format	The raw data format used to store tabular text data before analysis.

Implementation & Repository Structure

Repository Structure

```
Retail-Sales-Performance-Analysis/
│
├── data.csv
├── analysis.ipynb
├── readme.md
├── requirment.txt
└── summary.pdf
```

Execution & Data Processing Summary

Pipeline Stage	Implementation Details	Status
Data Ingestion	Loaded 1,545 transaction rows from `data.csv` using Pandas.	**Completed**
Data Cleaning	Dropped 45 duplicate rows, filled 78 missing prices using product median price.	**Completed**
Field Engineering	Added `Total_Sales` (Quantity × Unit_Price) and extracted `Transaction_Month`.	**Completed**
Core Visualizations	Generated 4 essential charts: Monthly Trend, Product Revenue, Category Sales, Location Sales.	**Completed**

Summary of Calculated Store Metrics

- **Total Store Revenue:** ₹1,101,336.28
- **Total Completed Orders:** 1,500 transactions
- **Total Physical Units Sold:** 3,682 items
- **Average Order Spend:** ₹734.22 per purchase

Key Findings & Business Recommendations

1. Key Analytical Findings

- **Sales Trends Over Time:** Store revenue remained strong across Q1 and Q2, with peak sales in **May (₹152,812)** and **June (₹146,898)**. Weekly shopping traffic was steady across all business days.
- **Best & Worst Selling Products:** The **Pro Gaming Laptop** is the single largest revenue driver, generating **₹653,895** (over **59%** of total store revenue) due to its high unit price. Items like **Plant Protein Powder (₹15,995)** and **Vitamin C Serum (₹20,340)** generated lower revenue despite healthy unit volumes.
- **Category & Store Performance: Electronics** leads all categories with **67.7%** of total revenue. Across city branches, **Hyderabad (₹249,036)** and **Chennai (₹228,471)** are top

performers, while Bangalore recorded the lowest sales.

2. Practical Business Recommendations

1. **Protect and Stock High-Value Inventory:** Ensure high-demand electronics like laptops remain safely stored and consistently in stock, as they generate the vast majority of revenue.
2. **Introduce Product Bundles:** Pair lower-priced, high-volume items (skincare serums and health supplements) with apparel or tech accessories to increase average order spend.
3. **Targeted Regional Promotions:** Run localized discounts and marketing campaigns in Bangalore and Delhi to help those stores reach the revenue levels of Hyderabad and Chennai.

3. Project Conclusion

All project objectives for **DA-011** have been completed. Raw transactions were cleaned, key store KPIs were calculated, and the four essential visual charts were generated. The findings provide clear, practical guidance for retail inventory management and sales strategy.

Project Links

Type	Link
GitHub Repository	GitHub Repo
Jupyter Notebook	analysis.ipynb
Raw Dataset	data.csv
Colab Notebook Link	Analysis Notebook

References

Technology	Reference
Python	Python Official Documentation
Jupyter Notebook	Jupyter Documentation
Google Colab	Google Colab Portal
Pandas	Pandas PyPI / Documentation
NumPy	NumPy Documentation
Matplotlib	Matplotlib Documentation
Seaborn	Seaborn Documentation
CSV	Python CSV Library Documentation

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